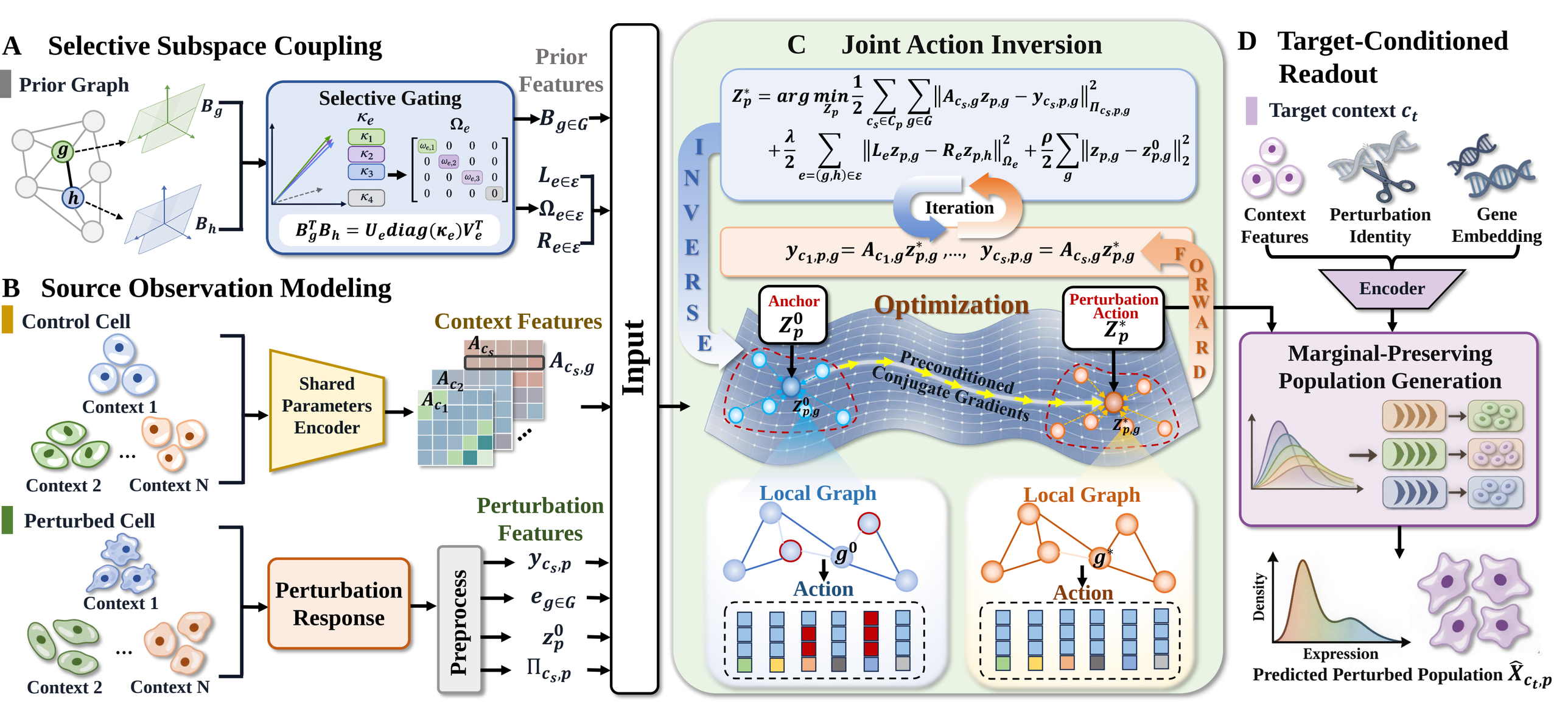

A Selective Subspace Coupling
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Selective Gating
Prior Features
B Source Observation Modeling
Control Cell
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Shared Parameters Encoder
Context Features
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C Joint Action Inversion
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D Target-Conditioned Readout
Target context c_t
Context Features
Perturbation Identity
Gene Embedding
Encoder
Marginal-Preserving Population Generation
Density
Expression
Predicted Perturbed Population $\widehat{X}_{c_t,p}$